Experiments with Thermal Comfort Analysis

System Documentation & Report

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Contents

1 Overview **4**
- 1.1 Motivation . . . 4
- 1.2 Objectives . . . 4

2 Temperature Measurement **5**
- 2.1 Ambient Temperature . . . 5
- 2.2 Wet Bulb Globe Temperature . . . 5
- 2.3 Mean Radiant Temperature . . . 6
- 2.4 Guidelines for Temperature Measurement & Instrumentation . . . 6

3 Relative Humidity Measurement **7**
- 3.1 Importance of Relative Humidity Measurement . . . 7
- 3.2 Choice of Sensor . . . 7

4 Survey into existing Comfort Control Models **8**
- 4.1 Existing Literature . . . 8
- 4.2 Analysis and Intepretation of existing models . . . 8
- 4.3 Propositions to Improve existing model . . . 9
  - 4.3.1 Dynamic Weights to adjust Temperature Range . . . 9
  - 4.3.2 Incorporation of Relative Humidity . . . 9
  - 4.3.3 Introduction of variation in the optimum indoor temperature . . . 9

5 Hardware Specifications **10**
- 5.1 Insights into System Design . . . 10
- 5.2 PCB Design . . . 10
- 5.3 PCB Design Guildlines . . . 10
- 5.4 PCB Manufacturing Specifcation . . . 12

6 Software Specifications **13**
- 6.1 Interfacing with DS18B20 Sensor . . . 13
- 6.2 Interfacing with DHT22 Sensor . . . 13
- 6.3 Implementation of Low Battery Indication . . . 13
- 6.4 Status LEDs . . . 13
- 6.5 Source Code Used to Record Readings . . . 14
- 6.6 Source Code to be implemented on ESP32 WSN . . . 16

7 System Testing **19**
- 7.1 Battery Charging . . . 19
- 7.2 Relative Humidity Measurement . . . 19
- 7.3 Temperature Measurement . . . 19
- 7.4 Battery Charging Indication . . . 19
- 7.5 Status LEDs . . . 19

8 Experimentation Methodology **20**
- 8.1 Measurement Accuracy and Comparison . . . 20
- 8.2 Installation of node in a classroom . . . 20
- 8.3 Setting up the Server Backend . . . 20

9 Observations **21**
9.1 Ambient Room Monitoring 21
9.2 Correlation of Humidity with Height 26
9.3 Globe Temperature vs. Ambient Temperature 28

10 Insights **31**
10.1 Designed Sensor Node vs. Classroom Installation 31
10.2 The correlation of RH with height 31
10.3 Globe Temperature Measurement 31

11 Conclusion **31**

12 Future Scope **32**
12.1 Experiments which can be tried out 32
12.2 Integration with Thermal Control Appliances 32
12.3 Improvements in Thermal Control Law 32
12.4 Exploring better batteries and connectivity capability 32

1 Overview

1.1 Motivation

Thermal Comfort Insights help us plan the user comfort in a controlled environment while using energy resources effectively. The problem introduces challenging elements such as data analytical modeling, precise and accurate instrumentation and cooperative control of thermal confort/ventilation appliances.

With a prototype design for a temperature & Relative Humidity sensing ESP32 Node, we plan ahead with an ambitious vision to devise a control law based on real-time data that we acquire from a thermal controlled environment.

1.2 Objectives

We plan to achieve the following in the course of this project:

- To understand different metrics related with temperature and propose a strategy which will be effective in monitoring user thermal comfort
- To explore & implement different temperature instrumentation techniques and apply them in an indoor atmosphere
- To find correlation between relative humidity & temperature and analyze the effect of relative humidity on user comfort
- To propose minor tweaks in existing thermal control laws and provide insights and justification for the same
- To complete the PCB design & testing of the ESP32 temperature nodes
- To install the nodes to capture data and verify/contradict claims made theoretically.

2 Temperature Measurement

2.1 Ambient Temperature

Ambient temperature is the temperature of the immediate surroundings in the room. This effectively translates to the temperature of the air medium around us in a room or outdoors.

This temperature can be regulated by the use of devices like air conditioner which regulate the temperature of the air it intakes and expels into the room. The temperature on the air conditioner remote is a long term set point to which the ambient temperature of the room will converge. However, the time it takes to reach that optimum threshold depends on the number of occupants in the room as well as the activity occurring in the room.

Based on the above discussion, ambient temperature can be thought of as a medium property of the air which participates in convective heat transfer.

2.2 Wet Bulb Globe Temperature

Wet Bulb Globe Temperature (WBGT) accounts for thermal stress in the atmosphere. This quantity is dependent on the amount of sunlight, the air temperature, wind speed, relative humidity and other sources of radiative heat in the vicinity. From a more philosophical stand-point the latent heat in the indoor environment can be measured using WBGT.

Using Wet Bulb Globe Temperature in the textbook fashion to assess thermal comfort would be an overkill. Instead we focus on measurement of Globe Temperature for our experiments and try to analyze the insights it provides us. Our focus is on capturing latent heat details which are missing in the ambient temperature readings.

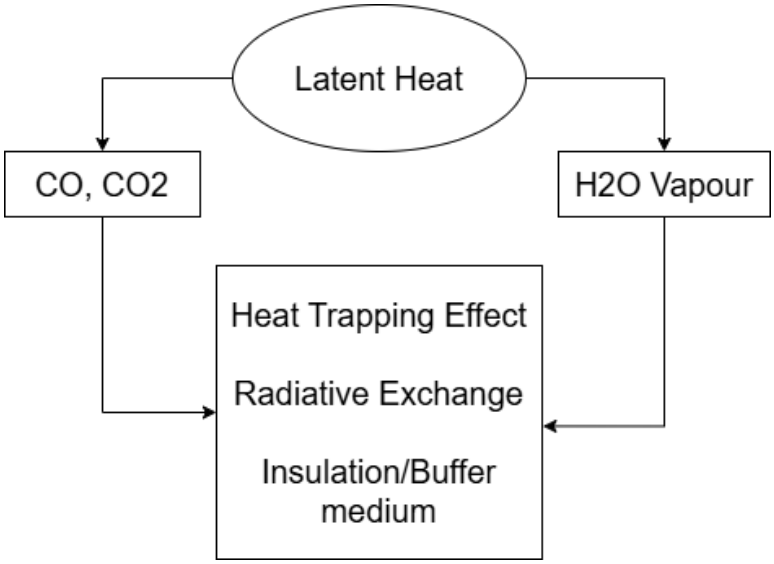

Figure 1: Manifestation of Latent Heat in Thermal Comfort Analysis

2.3 Mean Radiant Temperature

The Mean Radiant Temperature (MRT) is a measure used for evaluating the radiant heat environment near occupants and can be quantified as a measure of radiant heat exchange between occupants and their surroundings. MRT can be obtained using direct methods and indirect methods proposed in [3].

MRT requires air velocity modeling of indoor atmosphere, measurement of globe temperature and ambient temperature. As per [3] the accurate measurement of MRT remains an underexplored topic in occupied buildings.

Calculation and analysis of MRT for user thermal comfort would involve a large opportunity cost as it would require real-time modeling of air flow of a building and multiple spacial temperature measurements only to provide a sophisticated statistic. The integration of MRT into existing thermal comfort temperature law might not provide a great improvement compared to integrating Relative Humidity and Globe Temperatures into the control law.

2.4 Guidelines for Temperature Measurement & Instrumentation

Primarily 2 standards highlight the guidelines to be considered while measuring the physical quantities of an indoor environment, namely ISO 7726 (Ergonomics of the thermal environment - instruments for measuring physical quantities) and VDI 3787 (Methods for the Human Biometeorological Evaluation of Climate and Air Quality for Urban and Regional Planning at Regional Level)

These standards talk about different aspects of the subject as well as the system while measuring indoor physical quantities. The following is a brief summary of those points:

- The height and distance from the walls at which the reading are taken.
- The Stationarity of the readings taken shall dictate the measurement time interval.
- Understanding homogeneous and heterogeneous environments and making adequate appends to measure the characteristics of either environment.
- The accuracy and precision to consider measured data to be valid.
- Calibration techniques for different sensors.
- Air flow during ambient temperature measurement.

3 Relative Humidity Measurement

3.1 Importance of Relative Humidity Measurement

Relative Humidity (RH) by definition is the ratio of water vapour in air to the maximum capacity of water vapour that can be held by the air at the specific temperature, expressed as a percentage. Relative Humidity is an indication of the moisture we feel when we roam outdoors. As an psychological aspect of the problem, we do not feel comfortable if the relative humidity is towards the extremes.

Another way of looking at Relative Humidity (RH) is that it is a measure of the latent heat in the surrounding. Latent heat is the heat energy stored in a medium however doesn't show up directly as temperature during ambient temperature measurement. This heat is stored in the medium as the addition energy absorbed by water vapour to remain gaseous even at room temperature.

Latent heat is trapped by other green house gases as well such as Carbon Dioxide and other refrigerants. Quantities such as globe temperature and radiant temperature are able to expose latent heat through measurements and theoretical modeling.

3.2 Choice of Sensor

The accuracy bound on RH is general low and about $\pm 5\%$. We were faced with the choice of using a DHT-11 sensor (Accuracy: $\pm 5\%$) or its successor DHT-22 (Accuracy: $\pm 2\%$; Resolution: $\pm 0.1\%$). Both of these sensors use a similar protocol for communication and can be contrasted in the duration of the experiment as well. We planned to include the DHT-22 in our design for long term support privilege.

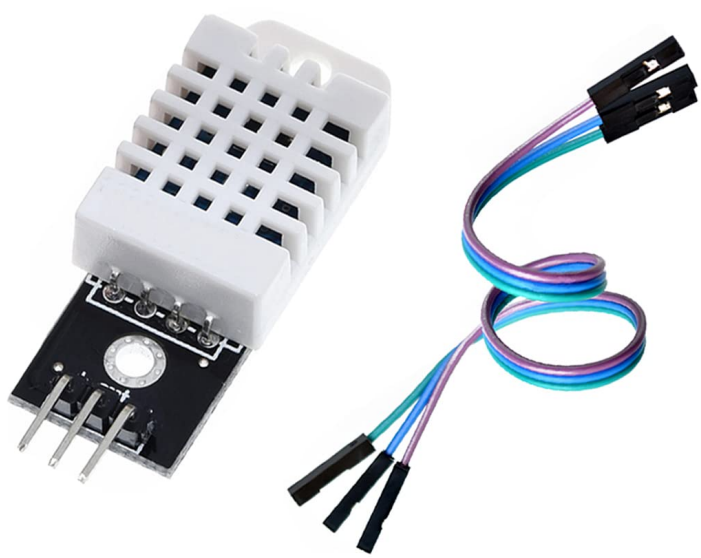

Figure 2: A DHT22 Temperature and Relative Humidity Sensor

4 Survey into existing Comfort Control Models

4.1 Existing Literature

One of the popular models we look at is the Indian Model for Adaptive Comfort (IMAC) [5, 6]. This is a model based of field surveys collected across 16 buildings in 3 seasons and 5 cities which represents five Indian climatic zones. The dataset used to build this model consists of 6330 responses gathered across various thermal control operating schemes such as naturally ventilated, mixed mode and air-conditioned buildings.

The study in [6] proposes an energy-efficient control strategy to provide thermal comfort to the occupants in an air-conditioned room. The claim made by the authors of [6] is that if we prioritize the usage of ceiling fan at an optimum set speed, we would be able to provide equal thermal comfort to the occupants with a higher temperature setpoint of the air conditioner as compared to when the air conditioner is running at a lower temperature setpoint. As a result, we are able to save on energy consumption. This study employs a machine learning model to identify the control law to be applied in order to maintain thermal comfort based on data collected from the occupants of the room.

The study in [6] identifies the importance of air movement over temperature control in warmer environments. It does so by incorporating an extension to the IMAC_upper temperature on the basis of ambient air speed in the room.

IMAC [5] comes up with the following bounds for indoor thermal comfort, also called the IMAC thermal comfort band.

$$IMAC_upper = ((0.28 \times outdoor\ temperature) + 17.87) + 3.46 \tag{1}$$

$$IMAC_lower = ((0.28 \times outdoor\ temperature) + 17.87) - 3.46 \tag{2}$$

The extension to the IMAC_upper range is provided using the following equation:

$$y = -1.39x^2 + 4.92x - 1.38 \tag{3}$$

where y is the shift in the upper band limit and x is the air speed.

The AC is only switched ON in case the temperature of the room exceeds the extended IMAC_upper temperature. The fan speed is varied when the temperature lies in the temperature range between IMAC_upper and extended IMAC_upper. The survey was conducted using a google form and thermal comfort was assessed on the basis of a public vote.

4.2 Analysis and Intepretation of existing models

We can breakdown 17.87 as a product of 0.72 and 24.82. Now if we study the inner bracket, it looks like a weighted average of 24.82^0C and the outdoor temperature. 24.82^0C can be thought of as a lower bound on the outdoor temperature for the duration of the study. To further clarify, 24.82^0C can be considered as an optimum indoor temperature for the local weather when the occupants of the room indulge sedentary activity with certain prescribed clothing for warm temperature zones.

The next component that we shall discuss is the weight used along with each temperature term. We observe a skew towards the indoor optimum temperature which is natural as we require cooler temperature in a restricted air flow environment.

The delta term of 3.46^0C can be interpret as a skew/variance/standard deviation which allows energy-efficient power switching while maintaining room temperature.

4.3 Propositions to Improve existing model

We can make the model more robust incorporating the following parameters:

4.3.1 Dynamic Weights to adjust Temperature Range

One obvious limitation to the proposed IMAC temperature range is the lack of temporal adaption in the temperature range. In view of having a simple model, this feature seems to be redundant and complicated to implement. However, incorporating temporal aspects can help us strike a better balance between energy consumption and thermal comfort.

4.3.2 Incorporation of Relative Humidity

Relative Humidity has a crutial role when it comes to thermal comfort. Water vapour being a greenhouse gas is capable of absorbing thermal energy which doesn't not directly reflect as ambient temperature. This is because such heat exchange is mainly a product of radiation based heat energy and cannot be measured by conductive/convective heat transfer. Thus in case of higher relative humidity thermal comfort can be prefered at a lower temperature. We can modify the IMAC temperature range based of this observation with a heuristic which is a direct function of measured Relative Humidity or the Relative Humidity difference between the outdoor and indoor environment.

4.3.3 Introduction of variation in the optimum indoor temperature

The thermal control model used in [5, 6] was built for a host of tropical environments and a subset of seasons in these environments. However, to make the model more robust we can vary the optimum temperature measure based on seasonal requirements. We will need a load of experiments to verify these claims but this could improve the performance of the model at the same average energy efficiency.

5 Hardware Specifications

5.1 Insights into System Design

2 configurations of sensor nodes were designed for the project. The first used a HDC1080 temperature sensor with an accuracy of $\pm 0.2^0 C$ and a DHT-11 temperature and relative humidity sensor to measure relative humidity with an accuracy of $\pm 5\%$.

A second configuration has two 3-wire ports which are used by DHT-22 temperature and relative humidity sensor and a DS18B20 temperature sensor which measures temperature with an accuracy of $\pm 0.5^0 C$.

The second solution happens to be a more cost effective solution to monitor Temperature and Relative Humidity and was thus prefered for production of PCB.

5.2 PCB Design

We have provided a schematic and PCB layout for the node we have designed. There were a few corrections/revisions were made post the testing of the manufactured PCBs when are now corrected in the attached files.

- The signal line of one of the ports was mapped to VP on the ESP32 which is now mapped to D25 which is an IO pin.
- The toggle switch was removed from the design as there is a provision in the TP4056 to intelligently connect the load and the charger to the battery as per requirement.
- The diode circuitry to indicate battery charging can be made robust.

We have optimized the design to a single side PCB for ease of manufacturing. Need for pullup resistors was not important for the one-wire protocol to function however, might be essential for an I^2C bus protocol with HDC1080 temperature sensor.

5.3 PCB Design Guildlines

- Routing Track Width = 0.35 mm
- Outer Diameter of Hole = 0.85 mm
- Hole width (diameter) = 0.7 mm
- Minimum Annular Width = 0.1 mm
- Copper to hole clearance = 0.25 mm
- Copper to edge clearance = 0.5 mm

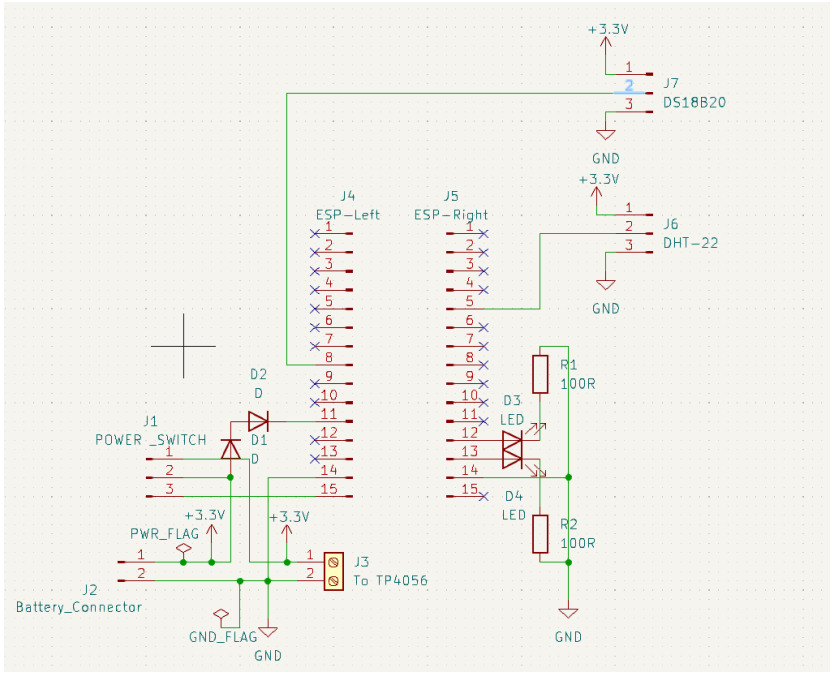

Figure 3: Schematics for the DS18B20 Sensor node

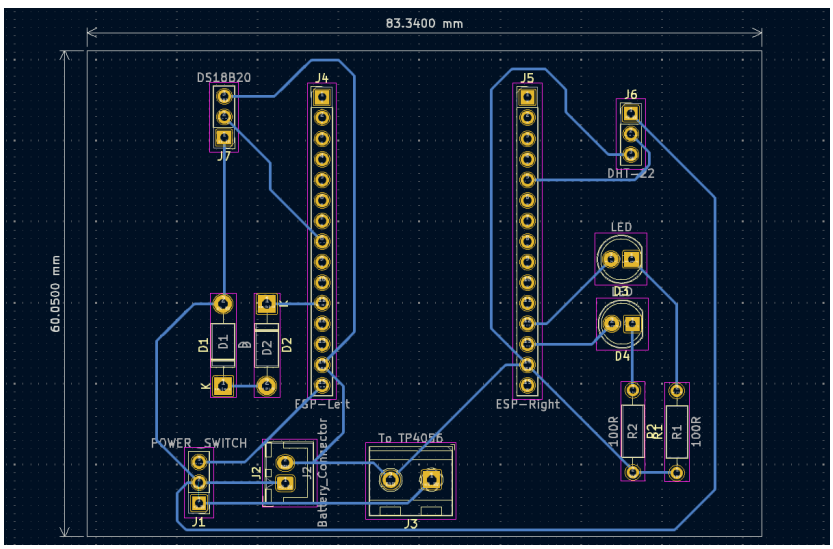

Figure 4: PCB Design of the manufactured PCB

5.4 PCB Manufacturing Specifcation

The PCBs were manufactured from PCB Lab, IIT Bombay. A silkscreen might be prefered for the final PCBs as it would increase the life of the sensor node circuitry. The required Gerber files can be found attached with this project.

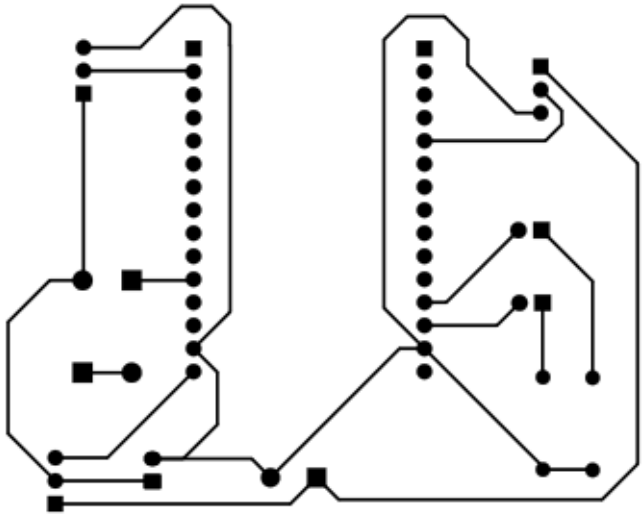

Figure 5: Gerber file layout of manufactured PCB

6 Software Specifications

6.1 Interfacing with DS18B20 Sensor

Please refer to the module documentation linked here.

6.2 Interfacing with DHT22 Sensor

Please refer to the module documentation linked here. Although this document is written with DHT11 in mind. The APIs used by the DHT22 RH & temp sensor are the exact same.

6.3 Implementation of Low Battery Indication

For the measurement of battery level we use the in-built ADC on ESP32 and set an indication once the voltage level sensed drops below a specific threshold.

The battery voltage level to be sensed is brought to the required voltage range using signal diodes. This might not be a good way to sense the battery level of the device as voltage drops may not be evident due to the internal circuitry of ESP32.

6.4 Status LEDs

The status LEDs connected to D2 and D15 on-board can be used for a variety of indications. They can be used to indicate requirement for charging of the battery. Or to run calibration programs on the node. These can be implemented as per the requirement of the experiment and need to be implemented.

6.5 Source Code Used to Record Readings

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include "DHT.h"

#define ONE_WIRE_BUS 25
#define DHTPIN 21
#define DHTTYPE DHT22
#define ANALOG_PIN 14

OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
DHT dht(DHTPIN, DHTTYPE);

int currentTime = 1344; // Renamed from 'time'
float DSB = 0;
float DSBavg = 0;
float DHTsum = 0; // Renamed to avoid confusion
float DHTavg = 0;
int i = 1;

void setup() {
  Serial.begin(115200);
  pinMode(2, OUTPUT);
  pinMode(15, OUTPUT);
  sensors.begin();
  dht.begin();
}

void loop() {

  digitalWrite(2, HIGH);
  delay(2000);
  digitalWrite(2, LOW);
  delay(2000);

  Serial.print("Time: ");
  Serial.println(currentTime);

  // Request temperature readings
  sensors.requestTemperatures();
  float dsbTemp = sensors.getTempCByIndex(0);
  Serial.print("Temperature by DS18B20: ");
  Serial.print(dsbTemp);
  Serial.println(" C");

  DSB += dsbTemp;
  DSBavg = DSB / i;
  Serial.print("DSBavg: ");
  Serial.println(DSBavg);

  // Read DHT sensor values
  float h = dht.readHumidity();
  float t = dht.readTemperature();

```

```
  Serial.println("DHT readings");
  Serial.print("Humidity: ");
  Serial.print(h);
  Serial.print("% Temperature: ");
  Serial.println(t);

  DHTsum += t;
  DHTavg = DHTsum / i;
  Serial.print("DHTavg: ");
  Serial.println(DHTavg);

  currentTime++; // Use renamed variable
  i++; // Increment counter

  digitalWrite(15, HIGH);
  delay(2000);
  digitalWrite(15, LOW);
  delay(60000); // Wait 60 seconds before next reading
}
```

6.6 Source Code to be implemented on ESP32 WSN

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include "DHT.h"
#include <WiFi.h>
#include "EspMQTTClient.h"

// Network credentials
const char* ssid = "Tanmay's Galaxy M52 5G";
const char* password = "qfdq2894";

// MQTT client
EspMQTTClient client(
  ssid,
  password,
  "broker.emqx.io",
  "emqx",
  "**********",    // MQTT password
  "TestClient",
  1883
);

// Pin definitions
#define ONE_WIRE_BUS 25
#define DHTPIN 21
#define DHTTYPE DHT22

#define LED1 2
#define LED2 15

// Deep sleep settings
#define Time_To_Sleep 60     // Sleep time in seconds
#define S_To_uS_Factor 1000000 // Conversion factor

// Sensor objects
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
DHT dht(DHTPIN, DHTTYPE);

RTC_DATA_ATTR int bootCount = 0;
int currentTime = 1500;

void onConnectionEstablished() {
  sensors.requestTemperatures();

  float avgtemp = 0.0, havg = 0.0, tavg = 0.0;
  for (int i = 0; i < 4; i++) {
    float dsbTemp = sensors.getTempCByIndex(0);
    avgtemp += dsbTemp;

    float h = dht.readHumidity();
    float t = dht.readTemperature();

    havg += h;
    tavg += t;
```

```
    delay(1000);
  }

  float dsbAvg = avgtemp / 4.0;
  float humAvg = havg / 4.0;
  float tempAvg = tavg / 4.0;

  Serial.println("Time: " + String(currentTime));
  Serial.print("DS18B20 Avg Temp: "); Serial.println(dsbAvg);
  Serial.print("DHT22 Avg Humidity: "); Serial.print(humAvg);
  Serial.print("% Temp: "); Serial.println(tempAvg);

  String payload = String(dsbAvg, 2) + " " + String(humAvg, 2);

  if (client.publish("/sensor_data", payload.c_str())) {
    digitalWrite(LED1, HIGH);
    Serial.println("Published: " + payload);
    delay(1000);
    digitalWrite(LED1, LOW);
  } else {
    Serial.println("Failed to publish data.");
  }

  WiFi.disconnect(true);
  delay(1000);

  digitalWrite(LED2, HIGH);
  delay(1000);
  digitalWrite(LED2, LOW);

  esp_sleep_enable_timer_wakeup(Time_To_Sleep * S_To_uS_Factor);
  Serial.println("Going to sleep for " + String(Time_To_Sleep) + " seconds...");
  esp_deep_sleep_start();
}

void setup() {
  Serial.begin(115200);
  delay(1000);

  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);

  sensors.begin();
  dht.begin();

  bootCount++;
  Serial.println("Boot number: " + String(bootCount));

  client.enableDebuggingMessages();
  client.enableHTTPWebUpdater();
  client.enableOTA();
  client.enableLastWillMessage("TestClient/lastwill", "I am going offline");
}

void loop() {
```

```
  client.loop();
}
```

7 System Testing

7.1 Battery Charging

The battery charging was done by connecting a 25W samsung C-type charger to the TP4056 C-type USB port on the node. The system charges reliably and has on-board red LED to indicate charging and a blue LED to convey that the battery is fully charged. For further details refer to the datasheet of the part.

7.2 Relative Humidity Measurement

We have compared the Relative Humidity measurement to the commercial installed Flamingo sensor-node system installed in a 50-seat LT classroom. The readings have been mentioned in the observation section.

7.3 Temperature Measurement

We have compared the Relative Humidity measurement to the commercial installed Flamingo sensor-node system installed in a 50-seat LT classroom. The readings have been mentioned in the observation section.

7.4 Battery Charging Indication

We had implemented a 2 diode drop ADC measurement of the live battery voltage; however, the strategy does not work in the device.

7.5 Status LEDs

The status LEDs were tested using a blink and sleep approach and work properly. We plan on using this to indicate the status of the sensor node in the future. These can be considered as available programmable indications.

8 Experimentation Methodology

8.1 Measurement Accuracy and Comparison

We have performed a series of temperature and RH measurements over a span of 15 minutes and compared it against the physical quantity measurement system preinstalled in these rooms.

We plan to interpret the globe temperature readings along with the ambient temperature readings as well. The data can be referred to in the observations and insights section of this report.

8.2 Installation of node in a classroom

We planned on installing these nodes in a classroom to autonomously collect data while the class is ongoing. We wish to test the battery life of our sensing nodes using this test methodology. However, the bottleneck is the institute network which doesn't allow these nodes to connect to it.

Such experiments can be later performed in the dummy classroom setups present in the institute which might have a separate hotspot network other than the institute internet.

8.3 Setting up the Server Backend

In the last revision of the system we had setup a survey system online which used 3rd party QR code generation websites to link to a website hosted on an online cloud platform. There should be an effort to move this system to a local backend server which can be trusted by the institute network. This final test would complete the installation of the nodes in place for an autonomous system.

9 Observations

9.1 Ambient Room Monitoring

We considered taking observations with our developed sensor node and comparing it to the readings taken using the sensing system installed in 50-seat LT classrooms. We have tabulated the results for a 15 minute experiment.

The Sensor node was arranged near the blackboard of the classroom which is about 1-1.5 meters below the device node D2. The sensors we set up so that they remained in the air not in contact with any solid body.

Timestamp	**Temp (DS18B20)**	**Temp (DHT22)**	**RH (DHT22)**
13:44	27.75^0C	30^0C	64.80%
13:45	27.75^0C	29.65^0C	66.20%
13:46	27.56^0C	29.37^0C	67.60%
13:47	27.42^0C	29.17^0C	68.50%
13:48	27.34^0C	29.08^0C	68.30%
13:49	27.28^0C	29.03^0C	67.80%
13:50	27.24^0C	29^0C	67.70%
13:51	27.20^0C	28.99^0C	67.50%
13:52	27.18^0C	28.98^0C	67.40%
13:53	27.16^0C	28.97^0C	67.40%
13:54	27.14^0C	28.96^0C	68.10%
13:55	27.14^0C	28.97^0C	68.60%

Table 1: Readings taken using the sensor node developed in the project

On the following page we have the recordings of temperature and humidity data of the system nodes installed in the 50-seat LT classrooms. There were 5 sensors attached in the room which provided data at different locations. The locations were Classroom front, back and centre, one near the AC and the last one near the window.
We can notice a few observations among the readings taken:

- The temperature readings taken near the blackboard using our ESP32 sensor node and the front device reading match within about a 1^0C.
- The relative humidity taken by the device roughly is equal to the average humidity of the classroom or lies within the range of humidity expected in the classroom.
- The maximum humidity readings are obtained near the AC which seems a little strange.

Figure 6: An image of the classroom with the installed sensor nodes

Timestamp	**Temperature (°C)**	**Humidity (%)**
13:45:43	33.48	48.27
13:46:44	33.52	48.46
13:47:44	33.55	48.38
13:48:45	33.53	48.38
13:49:46	33.49	48.24
13:50:46	33.53	48.30
13:53:48	33.46	47.88
13:56:51	33.26	47.21
13:57:51	33.29	47.36
13:58:50	33.38	47.22
13:59:50	33.42	47.20
14:00:51	33.39	47.17
14:02:53	33.41	47.58
14:03:53	33.42	47.39
14:04:54	33.51	47.45

Table 2: Device D1 (Ambient) Measurements

Timestamp	Temperature (°C)	Humidity (%)
13:45:49	29.09	76.21
13:46:49	29.19	76.21
13:47:50	29.40	76.18
13:48:51	29.42	75.80
13:49:52	29.40	76.12
13:50:52	29.28	75.88
13:52:54	29.52	76.01
13:53:54	29.22	75.67
13:56:57	28.75	76.63
13:57:57	28.56	77.11
13:58:56	28.46	77.14
13:59:56	28.70	77.01
14:00:57	27.99	77.12
14:01:58	27.33	77.10
14:02:59	26.76	76.66
14:03:59	26.32	76.12

Table 3: Device D2 (Front) Measurements

Timestamp	Temperature (°C)	Humidity (%)
13:45:55	28.49	84.49
13:46:56	28.42	84.42
13:47:56	28.53	84.43
13:48:57	28.57	84.30
13:49:58	28.59	84.35
13:50:58	28.36	84.31
13:53:00	28.56	84.52
13:54:00	28.49	84.29
13:55:01	28.79	84.38
13:56:02	29.04	84.41
13:57:03	28.74	84.57
13:58:03	28.76	84.58
13:59:02	29.17	84.56
14:00:02	28.48	84.76
14:01:03	28.05	84.60
14:03:05	26.77	84.49
14:04:05	26.20	83.72

Table 4: Device D3 (Near AC) Measurements

Timestamp	Temperature (°C)	Humidity (%)
13:45:00	28.37	57.02
13:46:01	28.34	56.95
13:47:02	28.25	57.17
13:48:02	28.37	57.42
13:49:03	28.48	57.24
13:50:04	28.31	57.24
13:52:05	28.43	57.23
13:53:06	28.27	56.98
13:54:06	28.31	56.90
13:55:07	28.04	58.01
13:57:08	27.74	59.63
13:59:08	27.62	60.10
14:03:11	26.12	57.90
14:04:11	25.71	57.53

Table 5: Device D4 (Back) Measurements

Timestamp	Temperature (°C)	Humidity (%)
13:45:06	28.63	57.34
13:47:07	28.71	57.47
13:50:09	28.71	57.28
13:51:10	28.70	57.29
13:53:12	28.85	57.27
13:54:12	28.74	57.24
13:55:13	28.53	57.43
13:56:14	28.26	58.52
13:57:14	28.18	59.37
13:59:13	28.01	59.72
14:01:15	27.22	59.67
14:02:16	26.62	59.72
14:03:17	25.82	59.51
14:04:17	25.17	58.64

Table 6: Device D5 (Near Window) Measurements

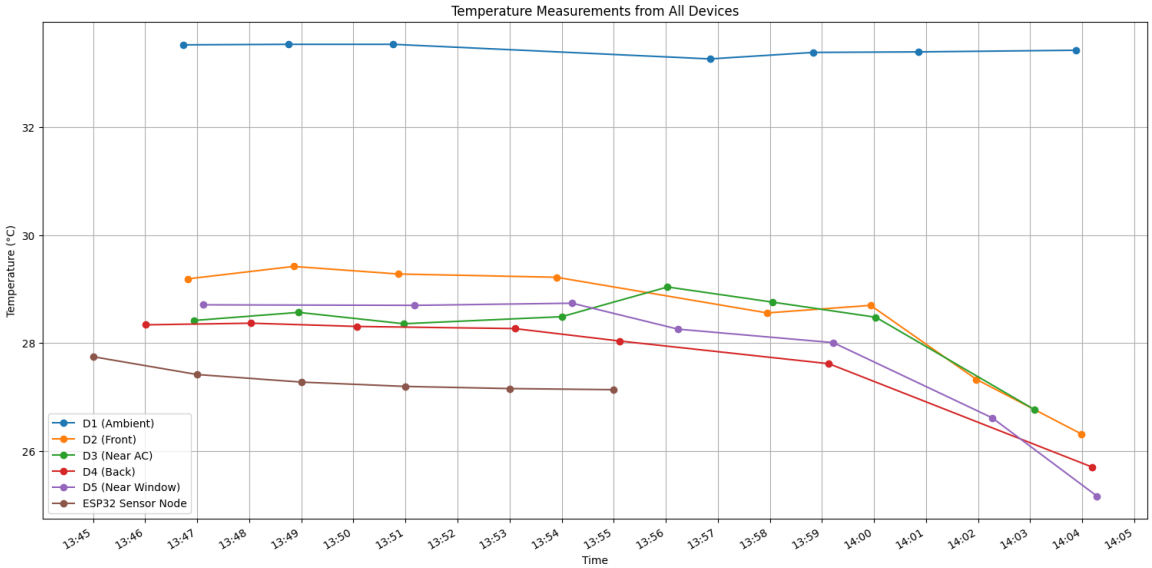

Figure 7: A plot of temperature readings obtained by each sensor on a running timeline

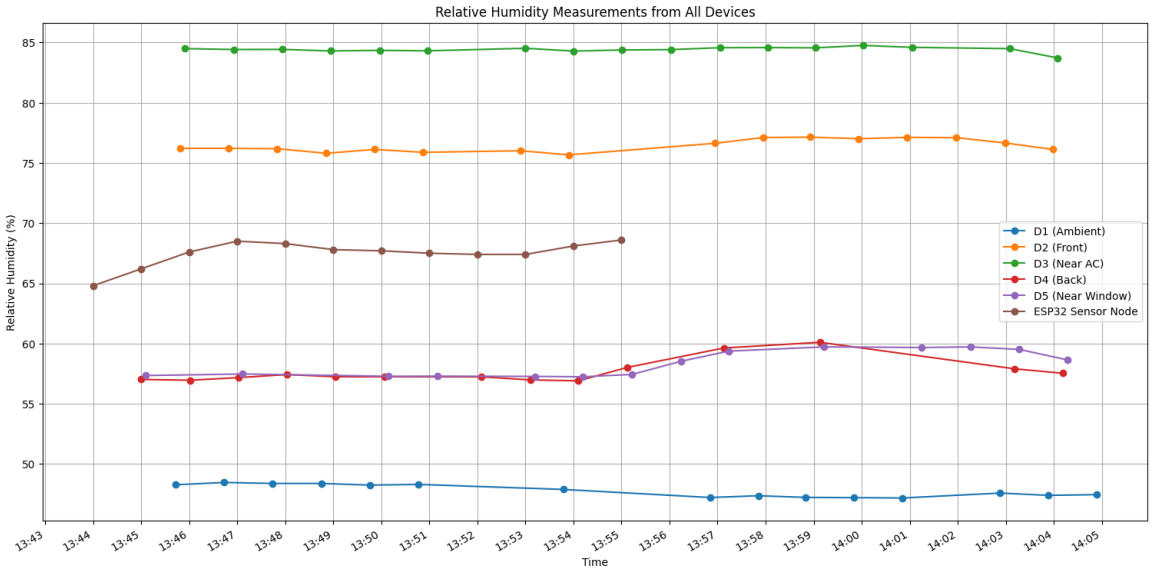

Figure 8: A plot of RH readings obtained by each sensor on a running timeline

9.2 Correlation of Humidity with Height

To understand whether there is a correlation between relative humidity and height at which it is measured in an enclosed room, we consider 3 junctions in along a vertical axis in a room and measure the RH & temperature for 5 minutes with 4 readings taken every minute to denoise the samples taken. The reason for recording temperature as well is to understand any correlation between the temperature and RH as well. This is done because we cannot take the readings at all 3 points at the same time. Thus, we have the temperature as an indicator for the physical quantity dynamic in the room.

We measure the RH & temperature at 3 heights 0.1m, 1m and 2m from the floor level.

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:01	26.97	27.90	73.40
15:02	27.03	28.00	73.80
15:03	26.98	28.10	73.95
15:04	26.91	28.10	73.65
15:05	26.83	28.00	72.95

Table 7: RH & temperature measurements at floor Height ($\tilde{0}$.1m)

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:08	26.89	27.80	73.55
15:09	26.87	27.95	73.80
15:10	26.92	28.00	73.95
15:11	26.89	28.00	73.55
15:12	26.80	28.00	72.80

Table 8: RH & temperature measurements at Table Height ($\tilde{1}$m)

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:18	26.87	27.90	73.85
15:19	26.92	28.10	73.75
15:20	26.80	28.10	72.90
15:21	26.70	28.10	72.10
15:22	26.59	28.00	71.70

Table 9: RH & temperature measurements at Top-Shelf Height ($\tilde{2}$m)

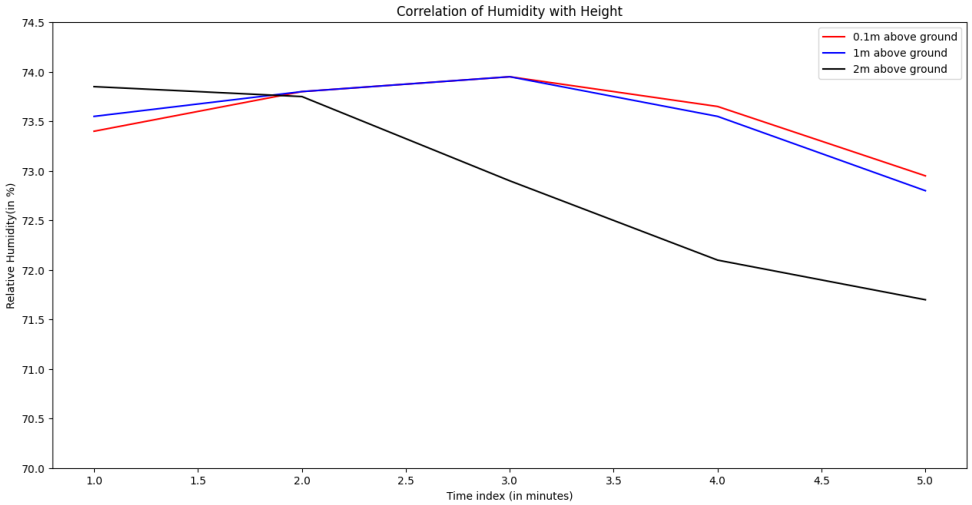

Figure 9: Comparison of Relative Humidity trends captured at different heights along the same vertical

We observe that there is not much variant of relative humidity measurement between the 0.1m and 1m plot lines. There is a small deviation in the 2m RH trend.

Although the readings taken are for a really small duration to make any meaningful conclusion, we can qualitatively divide the air volume in the room as below breathing level and above breathing level, centered around the human sources responsible for alteration of the air composition of the room.

9.3 Globe Temperature vs. Ambient Temperature

For recording this experiment we consider 3 setups, one with the DB18S20 sensor inserted in a white ping-pong ball, DB18S20 sensor inserted in a white ping-pong ball covered with black tape and finally the bare DB18S20 temperature sensor for ambient temperature measurement.

Each experiment lasts 5 minutes long with 4 samples taken every minute and averaged so as to apply a low pass filter on noisy sensor measurement model.

Below are the readings taken in a air conditioned room with 3 ceiling fans and 2-4 occupants.

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:33	26.72	27.80	70.90
15:34	26.73	27.80	71.40
15:35	26.84	27.90	71.90
15:36	26.92	28.00	72.30
15:37	26.94	28.00	72.25

Table 10: Ambient RH & temperature measurements

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:28	27.08	28.00	70.70
15:29	27.08	28.10	71.50
15:30	27.02	28.10	72.20
15:31	26.95	28.00	72.30
15:32	26.84	27.90	71.65

Table 11: White Globe temperature measurements along with readings from DHT22 Sensor

Timestamp	**Temp DB18S20 (°C)**	**Temp DHT22 (°C)**	**Humidity (%)**
15:38	27.55	27.95	71.50
15:39	27.52	27.80	70.80
15:40	27.16	27.80	70.60
15:41	26.97	27.90	71.35
15:42	26.89	27.80	72.40

Table 12: Black Globe temperature measurements along with readings from DHT22 Sensor

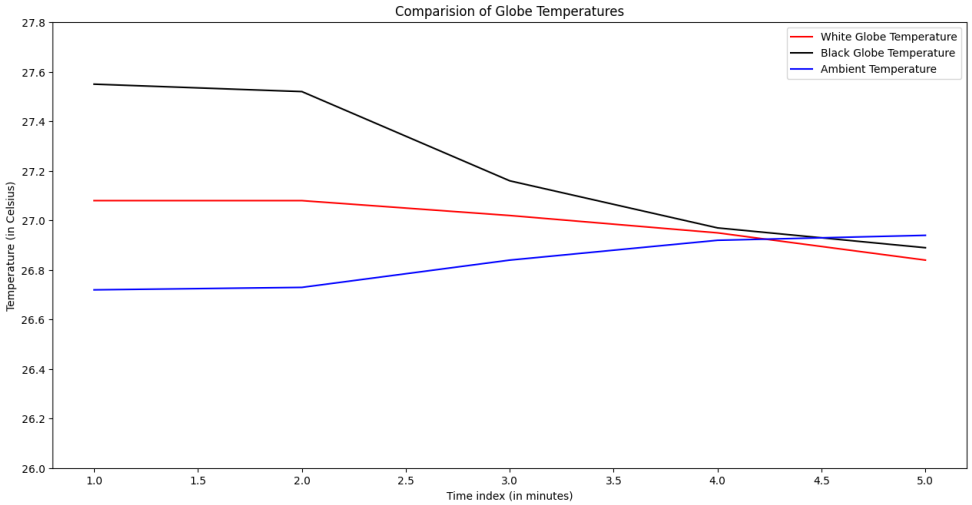

Figure 10: Comparison of globe temperature trends taken using the DB18S20 temperature sensor

We observe the globe temperatures to be about 1^0C higher than the ambient temperature which is a measure of the radiant temperature energy being captured in the temperature readings. There are more ways of capturing the globe temperature mentioned in the references of this report which can be tried.

Due to the unavailability of different material globes, we could not compare different globe temperatures. This can be tried as a part of future work where aluminium or iron spheres can be used to suspent the DB18S20 temperature sensors and record readings.

Figure 11: White globe setup

10 Insights

10.1 Designed Sensor Node vs. Classroom Installation

The temperature readings taken by the DB18S20 are similar to the temperature recorded by the D2 device which was the closed device to the ESP32 sensor node in the room. The temporal trend of temperature in the room is followed by the DB18S20 sensor. The Readings of the DHT22 sensor are about $1^0C - 1.5^0C$ above the readings recorded by the DB18S20 for all the experiments performed.

The relative humidity measured by the DHT22 sensor is within the range recorded by the other sensors in the room. And the values recorded seem plausible with the variation of RH in the room.

10.2 The correlation of RH with height

We observe that there is not much variation in RH at different heights in the room along the same vertical when the room is well ventilated with a ceiling fan. The effect of RH with height in a room without a ceiling fan can be studied further however the current conclusion seems that the relative humidity and temperature do not vary much with change in height within an enclosed room.

10.3 Globe Temperature Measurement

We observe that the globe temperature recorded by the sensor node is higher than the ambient air temperature which was expected as we wished to capture radient temperature with this setup. We use the DHT22 to get an idea of the ambient temperature at the time of measurement as this sensor is suspended freely without any enclosure on it.

We observe a difference of $\approx 1^0C$ between the ambient temperature and black globe temperature when an air conditioned room has 2-3 occupants.

11 Conclusion

To be certain with any instrumentation data we require to run such nodes continuously at least over a season. Using this System Report & Documentation we are certain that the nodes will work and record the temperature reliably.

We were able to successfully design a working PCB for sensor data acquisition and test multiple copies of the design. Minor rectifications proposed during the duration of the experiment have been updated in the PCB design for fresh off-the-box manufacturing.

The battery level indication is a subsystem that we were unable to complete and failed to detect volatage drop successfully during testing. A new approach must be used to design this feature.

12 Future Scope

12.1 Experiments which can be tried out

There are a series of experiments which can be run using the sensor node network which will help us understand temperature and RH variation of a room better in real time. We can study the effect of measurement of humidity at different heights in a room and interpret it in terms of relative weights of the component gases in the atmosphere.
We can study the correlation between measured temperature and RH and understand the trend it follows.

12.2 Integration with Thermal Control Appliances

We have methods to gauge the relative humidity as well as the temperature of the room. With augmentations prescribed in the report we can measure globe temperature as well and account for radiative heat exchange. Using this data and prescribed control laws we can improve the energy efficiency of controlled temperature environments.

12.3 Improvements in Thermal Control Law

Inclusion of user input feedback as well as physical quantity measurement and interpretation will help us gather better insights about the room under observation and the switching of temperature control devices can be planned for a long horizon rather than brisk switching which air conditioning load is not meant for.

12.4 Exploring better batteries and connectivity capability

In case of implementation of a smart home scheme from scratch, Power over Ethernet (PoE) and wired network capability can be explored to keep the sensor intervention concealed within the interior design of the room.

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